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Hepatic Lipoprotein Production, Cardiometabolic Phenotypes and Subtypes of Steatotic Liver Disease

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Abstract

Metabolic dysfunction associated steatotic liver disease (MASLD) impacts more than a third of adults worldwide. The burgeoning epidemic of MASLD has led to increases in atherosclerotic cardiovascular disease (CVD), which is the leading cause of morbidity and mortality among affected individuals, particularly those with type 2 diabetes. Subsets of patients with MASLD will develop progressive liver disease, including metabolic dysfunction associated steatohepatitis (MASH), advanced fibrosis, cirrhosis and hepatocellular cancer. The association of progressive forms of MASLD with cardiometabolic disease has spurred interest in understanding how distinct pathways in hepatic lipid and lipoprotein metabolism may identify candidate therapeutic targets to mitigate CVD risk. Deeper understanding of hepatic lipid and lipoprotein homeostasis in MASLD is a pressing need for several reasons connected with some of the molecular and signaling adaptations occurring in obesity and type 2 diabetes. First, because these conditions individually and in combination accelerate the onset and progression of MASLD; secondly because progressive MASLD may promote insulin resistance and type 2 diabetes and, thirdly because some CVD risk factors (dyslipidemia, obesity, type 2 diabetes) overlap with those for MASLD. Evidence suggests a multifactorial etiology for MASLD, including genetic factors that either promote or mitigate liver disease progression and which, along with environmental modifiers (diet, obesity) and insulin resistance, contribute to CVD risk. Segregating these etiologies of MASLD by phenotype/metabotype reveals a systemic metabolic dysfunction (obesity, insulin resistance) dominant subtype which is associated with increased CVD, and a second dominant subtype where familial and genetic factors predominate, and which reflects impaired VLDL secretion and is associated with reduced CVD risk. Further study of these distinctive subtypes of MASLD may identify new, tailored approaches to prevent cardiometabolic disease.

Subject Terms

Cardiovascular Disease; Lipids and Cholesterol; Diabetes; Type 2

INTRODUCTION AND OVERVIEW

Metabolic dysfunction associated steatotic liver disease (MASLD) is increasingly recognized as a global problem, with recent estimates suggesting a worldwide prevalence of 38%^{1,2}. The burden of MASLD is unevenly distributed, with highest prevalence observed

in people from Kuwait, Egypt and Qatar ³ and the highest rates of increase (2010-2021) observed in the most populated countries, namely China and India ³. Understanding of the geographic variation in MASLD prevalence has been advanced by studies showing the importance of a common genetic variant (*PNPLA3* rs 738409), which is predictive of both MASLD prevalence and highly correlated with disease progression to cirrhosis (reviewed in ⁴) and confirmed across different populations ⁵. Some racial and ethnic features of MASLD distribution were examined in the Dallas Heart study population where Hispanic individuals (~49% harboring the *PNPLA3* variant) exhibited the greatest prevalence (compared to European whites and Blacks) of hepatic steatosis ⁶. However, even after adjusting for the prevalence *PNPLA3* and other variants, non-Hispanic Black subjects appeared protected from MASLD ⁶. Of further interest, the rate of progression of chronic liver diseases such as hepatitis C was not statistically different in non-Hispanic Blacks ⁷, suggesting that the distinctive protection against MASLD (versus other forms of chronic liver disease) in Black subjects may reflect other factors, which will require validation. Those unexplained, racial and ethnic trends in the population distribution of MASLD highlight key unmet needs which will be explored below. These unmet needs include improved understanding of how underlying genetic variants interact with environmental and socioeconomic factors to influence MASLD development and progression. A related unmet need is to understand how the underlying mechanistic drivers promoting hepatic steatosis contribute, both individually and in combination, to the development of a full range of cardiometabolic comorbidities in MASLD/MASH, including CVD (as reviewed in ⁸). A central theme of this review is to highlight pathogenically distinct subtypes of MASLD. One subtype reflects steatotic liver disease in which substrate excess, obesity and insulin resistance dominates and is associated with dyslipidemia and increased CVD risk. A second subtype reflects a relative impairment of hepatic triglyceride (VLDL) export and is associated with steatotic liver disease, but with reduced circulating levels of atherogenic lipoproteins and reduced CVD risk.

Driven in part by the recognition that MASLD encompasses a spectrum of disease that include steatotic liver disease, but with the highest prevalence in people with obesity ⁹ and those with type2 diabetes ², the definition of MASLD now includes steatotic liver disease in the absence of at-risk alcohol consumption or other etiologies ¹⁰, and includes at least one of the comorbidities associated with insulin resistance (overweight/obesity, low levels of serum high density lipoprotein, increased serum triglyceride levels, hypertension or fasting hyperglycemia) ¹⁰. In very general terms, steatotic liver disease arises under conditions of nutrient excess (high saturated fat, glucose and fructose), which promote hepatic de-novo lipogenesis (DNL) and insulin resistance (reviewed in ¹¹). In the background of genetic variants that promote lipid droplet accumulation, homeostatic mechanisms (including VLDL secretion and mitochondrial function) in hepatic function become impaired, resulting in lipotoxic injury, inflammation, and progression to metabolic dysfunction associated steatohepatitis (MASH) ¹². Beyond the continuum of steatotic liver disease (MASLD/MASH) and the entwined relationship to obesity and type 2 diabetes there is also increasing recognition that alcohol intake may be a key environmental modifier, which has led to a revised nomenclature that now includes alcohol-related liver disease (Met-ALD) ¹⁰. This provision is important because the classification of patients within subcategories of MASLD appears to be dynamic and subject to change, as evidenced by

studies of over 1000 patients followed for more than 2 years, where 38% of subjects initially diagnosed with MASLD changed category between baseline enrollment and follow up and over 60% of participants with baseline Met-ALD changed categories ¹³. These findings highlight challenges in longitudinal studies of patients with MASLD, where variations in alcohol use may result in misclassification of subjects.

MASLD and Type 2 Diabetes: Causality or association?

There is continued debate in understanding how the liver adapts to systemic metabolic challenges because, while obesity and type 2 diabetes both initiate and promote hepatic steatosis, studies have suggested a reverse causality (as evidenced in Mendelian randomization) wherein hepatic steatosis, particularly when combined with fibrosis, promotes the development of insulin resistance, particularly in subgroups with known genetic predisposition ¹⁴. Those studies suggest that insulin resistance is an indirect consequence of hepatic steatosis, presumably mediated by fibrogenic activation signals and progressive liver damage ¹⁴. A recent review suggested that MASLD was an intrinsic cause of insulin resistance ¹⁵, citing studies showing that 71% subjects with biopsy-proven NASH and 46% subjects with NAFLD from a Swedish cohort developed type 2 diabetes over 13.7 years of follow-up ¹⁶. However, there was no baseline evaluation of glucose intolerance in those studies, making it difficult to conclude that hepatic steatosis itself was causative ¹⁶. Another study reported outcomes in over 25,000 Korean men, where 17.8% of subjects with moderate-to-severe NAFLD developed type 2 diabetes over five years of follow up ¹⁷. However, the diagnosis of NAFLD in that study was based on ultrasound examination alone and all subjects exhibited baseline increases in HOMA-IR ¹⁷, suggesting that progressive liver disease (ie inflammation and fibrosis) may exacerbate underlying insulin resistance and promote development of type 2 diabetes, reservations echoed in other reports (see ¹⁸). Other studies have demonstrated that hepatic insulin resistance can be induced by lipid droplet accumulation, potentially through pathways involving membrane-bound diacylglycerol-mediated alterations in protein kinase C epsilon ¹⁹. The countervailing view is that hepatic steatosis alone does not cause insulin resistance. Strong evidence in support of this view reflects findings ²⁰ in subjects where hepatic steatosis is caused by a genetic defect in hepatic export of triglyceride-rich lipoproteins (familial hypobetalipoproteinemia (FHBL), discussed in detail in a later section). Those studies showed that hepatic insulin sensitivity (measured by hyperinsulinemic-euglycemic clamping) was indistinguishable between FHBL subjects (with hepatic steatosis caused by impaired lipid export) and normal control subjects, matched for BMI ²⁰. Other studies showed that increases in hepatic steatosis are associated with progressive impairment of insulin action across liver, skeletal muscle and adipose tissue in non-diabetic obese subjects ²¹, suggesting that MASLD is associated with multiorgan dysfunction and impaired insulin sensitivity. Taken together, the available information in human subjects suggests that hepatic steatosis is associated with, but (at least in isolation) does not *cause*, insulin resistance and type 2 diabetes. Recent studies identified a panel of 235 circulating metabolites that, together with genetic and lifestyle risk factors, predicted future risk of type 2 diabetes over 26 years of follow up ²², raising the possibility that continued advances in metabolomic technologies across multiple racial and ethnic groups with steatotic liver disease may identify specific mediators promoting type 2 diabetes.

MASLD and Type 2 Diabetes: causal association with progression of liver disease and with cardiovascular disease

The coexistence of type 2 diabetes in patients with MASLD portends adverse outcomes, as shown in recent findings that patients with both conditions exhibited increased risk for decompensated liver disease and hepatocellular carcinoma. In a study of more than 2000 patients, and after adjustment for confounders, patients with type 2 diabetes exhibited increased risk of hepatic decompensation (sHR 2.15 [95% CI 1.39–3.34]; $p=0.0006$)²³. Furthermore, the presence of type 2 diabetes was an independent predictor of hepatocellular carcinoma development (sHR 5.34 [1.67–17.09]; $p=0.0048$), even after adjustment for baseline liver stiffness as a surrogate marker of hepatic fibrosis²³. Those findings strongly suggest that systemic mediators of metabolic dysfunction associated with insulin resistance and type 2 diabetes contribute directly to MASLD progression and the development of adverse outcomes.

Even though subjects with MASLD, particularly those with type 2 diabetes are at risk for developing advanced liver disease, CVD represents the major cause of mortality, with subjects at increased risk of atherosclerotic cardiovascular disease²⁴. A large meta-analysis reviewing studies reported over three decades up to 2019 found that studies where MASLD was confirmed by ultrasound, fatty liver index or by liver biopsy, revealed liver-specific mortality was increased 1.75 fold (95% CI 0.58–2.91) but cardiac-specific mortality was increased 5.54 fold (95% CI 2.72–8.35)¹. Another recent meta-analysis reporting 21 separate studies of CVD outcomes in adults with histologically confirmed MASH and MASLD reported odds ratios of CVD from 3.12 (95 % CI: 1.33–5.32) to 4.12 (95 % CI: 1.91–8.90). Coronary artery disease risk was increased in people with MASH in 6 of 7 studies, stroke risk was increased in 6 studies, and heart failure in 2 studies²⁵. In addition, retrospective studies from a Korean cohort demonstrated MASLD was associated with increased risk for myocardial infarction, independent of other risk factors²⁶. These findings reinforce the concept that CVD is prevalent in patients with MASLD and MASH yet leave unanswered the question of causality. Mendelian randomization studies suggested a causal relationship between MASLD and arterial stiffness, yet failed to demonstrate causality with heart failure or coronary artery disease²⁷ while recent studies, using Mendelian randomization in a genome-wide study including genetic variants associated with hepatic steatosis, concluded that the specific mechanisms by which liver fat accumulates, rather than the accumulation of hepatic lipid itself, is causally linked to MASLD outcomes, including CVD²⁸. An overarching theme of this review is to illuminate the contrasting roles of genetic and environmental modifiers in the cardiometabolic complications associated with MASLD.

Recognition of distinct MASLD phenotypes, both in humans and mouse models, has facilitated understanding of the role of genetic, dietary and other environmental factors in disease progression and the development of comorbid complications. However, the varying impacts of diet-induced MASLD in different mouse genetic backgrounds and the mechanisms and pathways modifying lipoprotein metabolism and cardiometabolic risk have yet to be fully explored as a surrogate for CVD susceptibility in humans with MASLD²⁹. Further understanding of how therapeutic modulation of hepatic and systemic lipoprotein metabolism may identify hepatic lipid homeostatic pathways that could be modified to

mitigate CVD, especially when paired with earlier disease detection³⁰ while acknowledging the need to distinguish association from causality³¹.

HEPATIC LIPOPROTEIN PRODUCTION AND CLEARANCE ARE ALTERED IN THE SETTING OF METABOLIC DYSFUNCTION AND INSULIN RESISTANCE IN MASLD: RELEVANCE TO CARDIOMETABOLIC COMPLICATIONS IN MASLD

Hepatocytes export neutral lipid (cholesterol esters and triglycerides) in VLDL in a multistep process of particle initiation, maturation, secretion. Neutral lipids are mobilized from endoplasmic reticulum (ER)-associated membranes and lipid droplets within the ER lumen^{32–36}, through fusion with APOB100, which is mediated by the heteromeric endoluminal chaperone, microsomal triglyceride transfer protein (MTTP)-protein disulfide isomerase (PDI) complex. The initial step in APOB100 lipidation occurs co-translationally, during elongation of the nascent APOB protein into the ER lumen³², an essential and rate-limiting step in VLDL assembly, as evidenced by subjects with rare variants in either *APOB* or *MTTP*, where VLDL secretion is virtually eliminated and plasma lipid levels and APOB100 concentrations are dramatically reduced, the resulting impairment in VLDL export causing progressive hepatic steatosis and features of MASLD/MASH³⁷. Under physiological conditions, APOB100 continues to acquire neutral lipid as the nascent VLDL particle progressively enlarges and undergoes transit through the tubulovesicular pathway, acquiring APOC3 and APOE which are incorporated into the mature VLDL particle (reviewed in³⁸), and secreted into the plasma as a source of triglyceride-rich lipoproteins (TRLs). Alterations in hepatic lipoprotein (VLDL) production and LDL clearance in subjects with MASLD and obesity/insulin resistance are summarized in FIGURE 1.

How does insulin resistance contribute to hepatic lipoprotein abnormalities in MASLD?

Insulin signaling plays a key role in regulating VLDL metabolism, physiologically promoting intracellular nascent APOB degradation and reducing VLDL secretion in both rodent models³⁹ and humans⁴⁰, and (under physiological conditions) suppresses gluconeogenesis and increases hepatic DNL, via activation of the SREBP pathway (reviewed in¹⁵), (FIGURE 1). In the setting of insulin resistance, lipoprotein particle (LDL) clearance becomes defective through systemic mechanisms including impaired lipoprotein lipase activation as well as direct action of hepatic-selective loss of insulin receptor signaling causing downregulation of LDL receptor expression⁴¹. The net effects of global insulin resistance thus impact both the secretion and clearance of APOB100-containing particles. The impact of insulin resistance on hepatic lipoprotein metabolism includes shifts in the number, size distribution, lipid composition, and clearance of particles across the VLDL-IDL-LDL spectrum⁴². Since serum triglycerides are transported predominantly within VLDL particles, hepatic triglyceride export capacity within VLDL is determined by the number of APOB100 particles secreted. APOB100 secretion itself is influenced by transcriptional regulation, protein degradation, lipid metabolism, and receptor-mediated clearance, each step of which is modified by insulin signaling⁴³ (FIGURE 1). In addition, the size distribution of newly secreted human VLDL is modified by hepatic neutral lipid

content, which is also subject to regulation by insulin through pathways including impaired lipid catabolism and increased DNL^{15,44}. Under physiological conditions, hepatic insulin signaling increases hepatic APOB100 degradation, decreases VLDL particle secretion, and promotes lipoprotein uptake⁴⁵. However, in the setting of insulin resistance, hepatic lipoprotein uptake is impaired⁴⁶, which results in accumulation of VLDL and remnants of triglyceride-rich lipoproteins (TRLs) which are considered important mediators of atherosclerotic CVD^{42,47–50} (FIGURE 1).

Despite more than a decade of research, the mechanisms driving the complex hepatic metabolic adaptations in insulin resistance are still debated¹⁵ and the pathways that promote hepatic lipid droplet accumulation are incompletely understood (FIGURE 1). In the setting of MASLD accompanied by insulin resistance (obesity, type 2 diabetes), insulin signaling is selectively impaired, physiological suppression of gluconeogenesis via AKT and FOXO1 signaling is attenuated yet hepatic glucose production is increased which, along with increased DNL, further exacerbates hepatic steatosis⁵¹. These adaptations have been variously referred to as the paradox of insulin signaling⁵² or selective hepatic insulin resistance⁵³. The mechanisms and mediators involved in orchestrating the gluconeogenic and lipogenic programs likely include signaling from other tissues¹⁵ (FIGURE 1).

There is also uncertainty regarding the mechanisms underlying the proatherogenic phenotype in the setting of MASLD and insulin resistance, particularly in relation to VLDL production and the role of the hepatic LDL receptor. A landmark paper in 2008 demonstrated that liver-specific insulin receptor deletion (LIRKO mice) exhibited an atherogenic lipid profile resulting from increased hepatic VLDL triglyceride and ApoB secretion, coupled with decreased LDL clearance associated with downregulation of LDL receptor expression⁴¹. When those (LIRKO) mice were fed an atherogenic high fat diet, they developed marked hypercholesterolemia and accelerated atherosclerosis⁴¹. Those studies established that hepatic insulin resistance alone (LIRKO mice were neither obese or hyperphagic and failed to demonstrate alterations in pathways of hepatic DNL) is sufficient to replicate atherogenic dyslipidemia and atherosclerosis susceptibility. However, other studies in mice with low levels of insulin receptor expression in the liver but lacking insulin receptor expression in peripheral tissues (*L1^{B6}*) crossed into *Ldlr^{-/-}* mice demonstrated unexpectedly reduced VLDL triglyceride and ApoB secretion and, when challenged with a western diet, exhibited reduced atherosclerosis⁵⁴. These authors also demonstrated that genetically obese (*ob/ob*) mice exhibited increased VLDL secretion, but they found (in contrast to Biddinger et al⁴¹) that siRNA knockdown of the liver insulin receptor decreased both VLDL triglyceride and ApoB secretion⁵⁴. These authors concluded that targeted reduction in hepatic insulin signaling might reduce VLDL production, but they also noted (as in the Biddinger et al study⁴¹) that insulin receptor targeting also reduced *Ldlr* expression which would be anticipated to adversely affect circulating levels of VLDL and LDL⁵⁴. These contrasting studies highlight another feature concerning hepatic insulin receptor function and VLDL secretion. Both the LIRKO mice and the *L1^{B6}Ldlr^{-/-}* mice exhibited gene expression features consistent with suppression of SREBP1 and DNL, yet insulin receptor silencing in *ob/ob* mice (which exhibit upregulation of DNL) decreased both VLDL triglyceride and ApoB production, suggesting that even in the setting of obesity and increased hepatic DNL, impaired insulin receptor signaling may limit

VLDL secretion^{41,54}. It bears emphasis that subjects harboring insulin receptor mutations manifest systemic insulin resistance yet are not obese and exhibit no hepatic steatosis, likely reflecting decreased DNL⁵⁵. Thus, fundamental issues remain to be resolved regarding the molecular mechanisms of altered hepatic VLDL APOB production in the setting of MASLD and hepatic (vs systemic) insulin resistance. Advances using cell- and tissue-specific gain and loss-of-function in preclinical models have highlighted key signaling pathways (as summarized above and FIGURE 1) where disrupted hepatic lipid and lipoprotein metabolism are altered by genetic/environmental modifiers, but those observations require validation in deeply phenotyped patient subsets, coupled with reductionist (liver slices, 3D organoids) models⁵⁶ and patient-derived induced pluripotent stem cells⁵⁷ in order to understand the relevance for patients with MASLD.

MECHANISTICALLY DEFINED METABOLIC SUBSETS OF PATIENTS WITH MASLD

(1). MASLD with systemic factor predominance

There is consensus that the combination of obesity and (generalized) insulin resistance, along with intake of diets enriched in high saturated fat along with high glucose and fructose-supplemented beverages, drives a shift to increase hepatic DNL that together represent a hallmark pathogenic signature in a metabolic subset of patients with MASLD (FIGURE 1, 2). This consensus arose from studies where hepatic DNL substrate flux was tracked in humans, originally in a small cohort of obese subjects with MASLD that showed elevated DNL in the fasting state that failed to increase postprandially, with 26% and 23% of liver triglyceride and serum VLDL triglyceride respectively derived from newly synthesized lipid⁵⁸. Other studies showed that obese subjects with increased hepatic steatosis exhibited greater release of palmitate from adipose tissue compared to subjects without hepatic steatosis, with both groups of subjects exhibiting increased VLDL triglyceride production rates without a change in APOB100 secretion⁵⁹. These findings substantiate a role for adipose tissue insulin resistance as a key contributor to the adaptations in hepatic VLDL production. More recent studies, again using metabolic flux to track VLDL kinetics, demonstrated that obese subjects with metabolic syndrome and elevated hepatic lipid content (magnetic resonance spectroscopy, ie MASLD) exhibited a 3.3-fold increase in hepatic triglyceride production (as VLDL) from DNL and a doubling of DNL in VLDL fatty acid content, compared to subjects without hepatic steatosis⁶⁰. These workers also confirmed that subjects with MASLD exhibited higher levels of nocturnal fatty acids and no suppression of the contribution from DNL with fasting⁶⁰. Further refinement of these conclusions came from recent studies where lean and obese subjects (both without hepatic steatosis) were compared to obese subjects with MASLD, where the contribution of hepatic DNL to intrahepatic triglyceride (palmitate) was 11% in lean subjects, rising to 19% in obese subjects without hepatic steatosis and 38% in obese subjects with MASLD⁶¹. These studies also showed that hepatic DNL was inversely correlated with hepatic (and whole body) insulin sensitivity and that weight loss decreased intrahepatic steatosis at least in part through decreasing DNL⁶¹. These observations collectively reinforce the concept that hepatic DNL is increased in the setting of obesity and insulin resistance and is accompanied by an increase VLDL-triglyceride secretion, which could be viewed as an adaptation to

limit hepatic steatosis (FIGURE 1). However, while hepatic DNL increases linearly with the initial increase in hepatic steatosis, there is likely a plateau effect where further increases in hepatic lipid are not accompanied by adaptive changes in VLDL triglyceride secretion to compensate for increased hepatic triglyceride production, which then exacerbates hepatic steatosis^{59,61}. Of further interest is the observation that exercise-induced weight loss led to a rapid decrease in hepatic steatosis but without changing VLDL triglyceride secretion, implying that hepatic lipid content itself is likely only an indirect driver of VLDL production⁶².

Therapeutic targeting of DNL in systemic factor predominant MASLD: The finding that patients with obesity and MASLD exhibit adaptive changes in hepatic DNL has bolstered the rationale for exploring therapeutic targeting of DNL in mitigating MASLD/MASH progression. Recent observations using heavy water to analyze DNL and hepatic triglyceride turnover in 123 subjects with MASH, including 20 subjects with cirrhosis, revealed that DNL remained elevated even in subjects with advanced fibrosis and cirrhosis and despite reduced hepatic lipid content in those subjects with advanced disease⁶³. Of further interest, hepatic DNL was positively correlated with both the number of VLDL particles and the abundance of APOB100, suggesting that upregulation of DNL may be a major driver of VLDL triglyceride and APOB100 production⁶³ (FIGURE 1). Finally, these authors showed that treatment with the acetyl-CoA carboxylase inhibitor firsocostat reduced DNL in subjects with MASH both with and without advanced fibrosis, suggesting that strategies to reduce hepatic steatosis are feasible even in the later stages of disease⁶³. In a different approach to pharmacologic modulation of DNL, a recent phase 2b (proof of concept) trial of definistat (an oral fatty acid synthase inhibitor) in patients with MASH and F2 or F3 fibrosis revealed MASH resolution without worsening of fibrosis, suggesting that strategies to reduce hepatic DNL are feasible although longer term studies across multiple countries will be required⁶⁴. Another phase 2b trial reported findings in 392 patients with MASH and advanced (F3 or F4) fibrosis treated with a combination of cilofexor (a nonsteroidal FXR agonist) and firsocostat for 48 weeks, which revealed significant reductions in steatosis, lobular inflammation and liver stiffness, suggesting that combination therapies targeting DNL and promoting FXR signaling may have benefits for fibrosis regression⁶⁵. On the other hand, several acetyl-CoA carboxylase inhibitors have been evaluated in clinical trials with concerns regarding increased plasma triglyceride levels⁶⁶, possibly mediated by SREBP1 activation. In addition, recent studies in mice where both isoforms of acetyl-CoA carboxylase were deleted in liver showed increased fasting gluconeogenesis from alternative substrates including pyruvate and amino acids⁶⁷. Those findings suggest a need to explore adaptive pathways of hepatic intermediary metabolism following long term pharmacologic acetyl-CoA carboxylase inhibition in humans, which may be different to those observed in mice⁶⁸. A recent summary of pharmacologic therapies that reduced hepatic steatosis in patients with MASH included GLP-1 agonists⁶⁹, with earlier studies demonstrating that these agonists reduced both hepatic VLDL production and DNL in the setting of insulin resistance in preclinical models⁷⁰. The widespread use of GLP-1 and newer incretin-based interventions will likely have a major impact on CVD risk in patients with MASLD⁷¹.

A further clinical subset of MASLD includes subjects who are not obese, the so-called 'lean MASLD', may represent up to 20% of subjects with steatotic liver disease ⁷². Observational studies from a cohort of 1339 biopsy-proven Caucasian subjects including 195 (14% total) lean (BMI <25) subjects revealed less severe histologic disease and no difference in overall mortality over a period of 94 months ⁷³. Of importance is that the lean MASLD group were significantly younger than the non-lean group and when long term outcomes were adjusted for age, there was no difference between the groups ⁷³. A more recent cross-sectional analysis of 312 prospectively enrolled patients compared lean subjects (40) with MASLD to 90 subjects who were overweight and 88 subjects with class II or greater obesity, all age and sex-matched ⁷⁴. Those findings revealed an indistinguishable prevalence of fibrosis in lean (27%) and non-lean (31%) subjects with MASLD, even among subjects with high genetic risk score prediction, suggesting that MASLD progression may be similar in lean and obese subjects despite differences in BMI ⁷⁴. Detailed understanding of the underlying pathophysiology in lean MASLD is incomplete, although increased hepatic DNL and alterations in visceral adipose metabolism are considered relevant pathways ^{11,75}. A small single center study of 20, predominantly female Chinese subjects with lean MASLD (BMI 22.6) compared to 20 control subjects (BMI 21.2) revealed higher circulating triglyceride levels and lower HDL along with alterations in serum fatty acid profiles suggestive of altered DNL ⁷⁶. A recent retrospective study of 67,519 patients from the TriNetX-based database with lean MASLD reported significantly increased risk of CVD (compared to non-lean patients with MASLD), including new onset heart failure, composite cardiovascular outcomes and all-cause mortality ⁷⁷. A study of European subjects with lean MASLD demonstrated elevated serum concentrations of isobutyrate, methionine sulfoxide, propionate and phosphatidylcholines, suggestive of enrichment with microbial-derived metabolites ⁷⁸. Those studies suggest that lean MASLD may represent yet another distinct metabolotype, particularly in relation to the role of the gut-liver axis. Further study of patients with lean MASLD, including from different racial and ethnic groups and socio-economic backgrounds and matched for genetic propensity, insulin resistance and circulating lipoprotein characterization will be needed to advance understanding of the propensity to cardiometabolic outcomes in this subset of patients with MASLD.

(2). MASLD with familial/genetic factor predominance

Advances in risk stratification for CVD in subjects with MASLD have been accomplished using lipidomic and metabolomic profiling. One such study examined more than 1100 patients from several centers in Europe, who were matched for age, sex, BMI and insulin resistance/type 2 diabetes, and all of whom had biopsy-confirmed MASLD or MASH ⁷⁹. These subjects underwent extensive metabolomic profiling along with nuclear magnetic resonance lipoprotein analysis, which revealed three distinct subsets ('metabotypes') whose profiles were aligned with metabolomic signatures obtained from mice models where VLDL triglyceride secretion was either upregulated or impaired ⁷⁹. Subjects with impaired VLDL secretion ('A metabolotype') also exhibited lower levels of serum APOB, triglyceride and cholesterol as well as lower levels of LDL and TRL particles, which together align with a profile of reduced CVD risk ⁷⁹. By contrast, subjects with the 'C metabolotype' exhibited increased (2-3 fold) VLDL secretion, increased APOB levels and an atherogenic profile indicating increased CVD risk, ⁷⁹. The distribution of metabotypes was independent of

lipid lowering medication use and unrelated to clinical indices of insulin resistance (HOMA-IR and HbA1c), suggesting that other mechanisms may account for these metabotypes. Although this study was underpowered to detect differences in genetic variants (discussed below), the findings demonstrate that, despite underlying similarities in MASLD phenotypes (across several centers) with respect to age, obesity, insulin resistance and medication use, it is possible to parse out defined metabolomic subsets of patients with different risk profiles for developing CVD ⁷⁹. These themes are summarized in FIGURE 2.

The findings above, summarizing clinically defined subsets of patients with MASLD, highlight important differences in the underlying pathogenesis and outcomes and there is further evidence that genetic variants influence key outcomes, including progressive liver and cardiovascular disease, in a manner that reflects nutritional and environmental factors ⁸⁰. Studies from a Finnish cohort examined obese subjects with MASLD who were categorized based on an aggregate genetic risk score that included five common variants (*PNPLA3* rs738409; *TM6SF2* rs58542926; *MBOAT7* rs641738; *HSD17B13* rs72613567; *MARC1* rs 2642438), whose functional impact will be described in a later section. Substrate flux and serum NMR metabolomics were examined, which revealed increased fatty acid, amino acid and glucose levels and features of increased DNL in a subset (Met-Comp) ⁸¹. By contrast, the subset with predominantly genetic variant enrichment (Gen-Comp) were characterized by an increased mitochondrial redox state (increased serum beta-hydroxybutyrate/acetoacetate ratio) and decreased DNL ⁸¹. This study was among the first to combine deep metabolomic profiling along with measures of DNL and to combine those analyses with a limited genetic risk profile, and highlighted the emerging concept that the pathogenesis of MASLD reflects a balance between substrate surplus and DNL and is modifiable by underlying genetic variants that influence energy utilization and lipid export ⁸¹. The concept of systemic factor predominant (akin to the Met-Comp subset above) and familial/genetic factor predominant (akin to the Gen-Comp subset above) subtypes is illustrated in FIGURE 2.

MASLD exhibits a strong genetic/inherited component, as shown in studies examining twins, which established that hepatic steatosis and fibrosis are heritable traits, particularly in monozygotic twins ⁸². Building on these and other studies alluded to above, recent studies using the UK Biobank showed that visceral adiposity, whole body fat mass and BMI were independent predictors of hepatic steatosis, inflammation and fibrosis and then used these adiposity markers to generate an adjusted genome wide association score which revealed 27 previously unassigned loci associated with MASLD, of which 6 were replicated in four independent cohorts ⁸³. They identified 37 independent loci associating with hepatic steatosis, including loci associated with regulation of plasma lipoprotein metabolism and further refined their analysis to develop a partitioned polygenic risk score (PPRS) that allowed them to allocate known and previously unidentified variants into clusters that associated with either (1) impaired VLDL secretion or (2) increased substrate flux, energy uptake or decreased beta oxidation ⁸³. They observed that the PPRS from cluster 1 was associated with decreased CVD while the PPRS from cluster 2 predicted increased CVD risk and they further observed that both PPRS groups were predisposed to develop type 2 diabetes, again reinforcing the concept that there are functional overlaps in the underlying pathways that predispose to MASLD ⁸³. As further reinforcement of the overlap concept, the

PNPLA3 variant rs738408 (pI148M), which is the strongest genetic variant predisposing to MASLD, appeared in both clusters⁸³, yet is not causally linked to ischemic heart disease⁸⁴. Thus, despite some overlap (as previously observed), those findings identified two distinct types of steatotic liver disease, one being liver centric (cluster 1), and the other (cluster 2) that reflects the complex adaptations associated with metabolic syndrome, obesity and type 2 diabetes. These clusters are represented schematically in FIGURE 2.

Clustering of MASLD subtypes into two distinct etiological groups was substantiated in a parallel approach using deep phenotyping based on large population datasets and focusing on cardiometabolic and hepatic traits which replicated the findings above by identifying a cluster of patients with 'liver-specific' MASLD, defined by genetic linkage and rapid progression of liver disease but reduced CVD and a second cluster that aligned with abnormal glucose metabolism, elevated circulating triglyceride levels, associated with both increased CVD risk and increased type 2 diabetes⁸⁵. A subset of patients had available liver transcriptomes, and another subset had available plasma metabolomes, each of which revealed distinctive profiles that segregated by cluster. These findings demonstrate that patients with MASLD and similar liver phenotypes (steatosis, fibrosis) can be segregated by divergent etiologies and with different clinical trajectories regarding CVD risk (FIGURE 2)⁸⁵. Both studies thus highlight the clinical and genetic heterogeneity of MASLD and add perspective on the future of informed precision approaches to patient management^{83,85}.

The above two studies were further substantiated in a genome-wide study in 37,358 participants from the UK Biobank where 13 genetic variants associating with hepatic steatosis were examined for effects on MASLD outcomes, including CVD²⁸. These 13 variants clustered into 3 groups: Group 1 (*TOR1B*, *MBOAT7*, *MARC1*, *GPAM*) were associated with lower levels of serum triglycerides and higher levels of LDL and HDL cholesterol, but no increased risk of CVD; Group 2 (*PNPLA3*, *TM6SF2*, *APOE*, *SUGPI*) were associated with lower serum triglyceride and LDL levels and reduced CVD (myocardial infarction) risk but increased risk of type 2 diabetes; Group 3 (*TRIB1*, *GCKR*, *ADH1B*, *CDHR4*) were associated with higher serum triglyceride and LDL cholesterol levels and a higher risk of CVD (myocardial infarction) but a lower risk of type 2 diabetes²⁸, (FIGURE 2). A central conclusion of that study was that hepatic steatosis itself is not likely to be causative in the association of MASLD and CVD; rather this association reflects the underlying, mechanistic etiology of MASLD²⁸. More recent studies identified 22 pathogenic or likely pathogenic variants in *APOB*, *MTTP*, *ANGPTL3* and *LDLR* among 24 of 3358 patients with MASLD, demonstrating that *APOB* variant carriers exhibited increased steatosis and a trend to increased fibrosis, along with reduced serum LDL and triglyceride levels⁸⁶. Those findings (summarized in FIGURE 2) suggest that monogenic disorders of dyslipidemia occur rarely in patients with MASLD but may be associated with more progressive disease⁸⁶.

Recent studies have highlighted the conclusion²⁸ that the *PNPLA3* variant is associated with reduced CVD, which is a relevant consideration in evaluating the therapeutic risk/benefit of siRNA targeting. One study used two step Mendelian randomization and liver fat quantitation to demonstrate that *PNPLA3* silencing increased levels of LDL-cholesterol, suggesting that decreasing or inhibiting *PNPLA3* might increase CVD risk⁸⁷. Those

findings were called into question by another study⁸⁸ that utilized the UK Biobank *PNPLA3* quantitative trait loci rather than liver fat to impute phenome-wide association, which demonstrated that patients harboring the variant *PNPLA3* did not exhibit reduced LDL cholesterol at baseline (in agreement with other observations⁸⁹) and suggest that therapeutic silencing of *PNPLA3* would not increase CVD risk⁸⁸. Some of those observations are further discussed below.

GENETIC SUBSETS OF PATIENTS WITH MASLD: UNDERLYING MECHANISMS AND THERAPEUTIC POTENTIAL

While it is beyond the scope of this review to cover all the genetic variants associated with MASLD, a summary of the functions/mechanisms associated with major variants linked to steatotic liver disease and progression and those associated with decreased circulating triglyceride and LDL cholesterol levels is relevant to an understanding of CVD risk as an outcome of MASLD. Recent work has emphasized that inherited risk factors are predictive of fibrosis development with age in patients with MASLD⁹⁰, but the mechanistic relationship of progressive steatotic liver disease and CVD merits additional study.

As noted above, the variant *PNPLA3 rs738409* encoding an I148M substitution mutant protein is the most prevalent and predictive determinant for MASLD^{4,5} (FIGURE 3) with more recent findings confirming a role in progression of MASLD to cirrhosis particularly in Caucasian and Asian populations⁹¹. Studies in 93 Finnish subjects demonstrated that carriers harboring the *PNPLA3* variant exhibited increased beta-hydroxybutyrate levels and reduced hepatic DNL after an overnight fast⁹². Those same patients demonstrated increased channeling of fatty acids towards ketogenesis after a mixed meal along with increased mitochondrial redox, and further demonstrated increased intrahepatic lipolysis and decreased hepatic mitochondrial citrate synthase flux after consuming a ketogenic diet⁹². These findings suggest that homozygous carriers of the I148M *PNPLA3* variant exhibit defects in fatty acid/substrate channeling for DNL as well as alterations in intrahepatic triglyceride turnover and defective mitochondrial function. However, obese patients harboring the *PNPLA3* variant were shown to be equally insulin resistant as non-carriers, at the level of liver, skeletal muscle and adipose tissue, suggesting that the underlying metabolic dysfunction in obese subjects stratified by *PNPLA3* genotype is related to the gain-of-function effects of the variant (reviewed below), rather than differences in insulin resistance⁹³. The finding of increased intrahepatic lipolysis in I148M variant *PNPLA3* carriers when fasting is consistent with decreased accumulation of the mutant protein on lipid droplets (*PNPLA3* is known to be insulin responsive) which then de-represses ATGL activity (as discussed below)⁹².

Recent studies explored the mechanisms underlying altered lipid droplet lipolysis in transgenic mice expressing the I148M *PNPLA3* variant, demonstrating that the lipid droplet associated protein ABHD5, which is a cofactor for ATGL, associates preferentially with *PNPLA3* and prevents activation of (ATGL)-mediated lipolysis⁹⁴. The I148M *PNPLA3* variant accumulates on lipid droplets in mice⁹⁵ (and humans harboring the variant), suggesting that increased expression of the mutant *Pnpla3*^{I148M} protein aggregate and

bind to ABHD5, although there was no difference in the affinity of WT versus I148M variant *PNPLA3* for binding to ABHD5⁹⁴. These findings demonstrate that the I148M variant *PNPLA3* causes a gain-of-function effect by sequestering ATGL and impairing lipolysis⁹⁴. Among the unanswered question from these studies is the impact of the I148M variant *PNPLA3* in regulating VLDL secretion. Findings from the Hobbs-Cohen lab showed hepatic steatosis with no alteration in VLDL production in mice expressing the *Pnpla3* variant^{96,97}, while studies from others showed that liver specific *Pnpla3* deletion or I148M knockin in mice fed a lipogenic diet resulted in decreased VLDL secretion⁹⁸. These divergent phenotypes in mice may reflect differences in the approaches and dietary conditions employed but leave open the possibility that the *PNPLA3* variant may modify hepatic VLDL production.

Finally, proof-of-concept studies support therapeutic knockdown of the mutant *PNPLA3* in liver as a strategy for reducing hepatic steatosis (reviewed in⁹⁹) and two recent phase 1 trials in humans harboring the I148M variant *PNPLA3*, using silencing siRNA⁸⁹ showed a durable (24 week) 30-40% reduction in hepatic steatosis, while another phase 1 trial in humans with MASH (harboring the I148M variant *PNPLA3*) treated with a GalNAc-conjugated antisense showed an 89% decreased in *PNPLA3* mRNA and protein on hepatic lipid droplets and resulted in a 12% decrease in liver fat, along with increasing the abundance of polyunsaturated fatty acids in plasma triglycerides¹⁰⁰. As noted above, a long-term risk of adverse cardiovascular outcomes in relation to changes in serum LDL cholesterol was raised in a study using two factor Mendelian randomization⁸⁷, but those concerns were refuted by another genetic causation study using Mendelian randomization, but focusing on *PNPLA3* linkage directly (rather than hepatic fat) to perform a phenome-wide association study⁸⁸. Those findings suggest that *PNPLA3* knockdown does not increase levels of LDL cholesterol or other atherogenic lipids.

Genetic variants associated with MASLD, impaired VLDL secretion and reduced CVD (Table 1 with additional references not discussed in text¹⁰¹⁻¹²⁴).

Population studies have highlighted the strong association between serum triglyceride levels and CVD and further suggest that triglyceride-rich lipoprotein particles (VLDL and VLDL-remnants) contribute to residual cardiovascular disease even in patients where LDL cholesterol levels are at target¹²⁵. Those findings, coupled with the demonstration that VLDL particle concentration (determined by NMR) was strongly and directly correlated with CVD risk⁵⁰, suggest that hepatic production of VLDL (triglyceride and APOB) may be a representative biomarker for CVD risk in subjects with high triglyceride levels¹²⁶. Further population studies concluded that triglyceride rich VLDL remnant particles account for about 4 times greater atherogenicity on a per-particle basis than LDL¹²⁷. These studies together affirm the importance of understanding how genetic variants that modify VLDL assembly and secretion modify CVD risk in patients with MASLD. Since VLDL particles contain one molecule of APOB per particle, along with varied neutral lipid cargo (cholesterol ester and triglyceride) and other associated polar lipids and proteins, it is imperative to understand how progressive steatotic liver disease alters VLDL lipid and protein composition and catabolism (to TRL remnant particles and LDL) to understand the impact on CVD risk. Loss-of-function variants shown to impair VLDL secretion include

APOB, MTTP and TM6SF2, genes whose roles include neutral lipid transfer into the ER lumen and assembly and maturation of intracellular VLDL particles (FIGURE 3 and reviewed in ¹²⁸). APOB is the structural protein responsible for accommodating the addition of neutral lipid whose transfer from the ER membrane bilayer and adjacent intraluminal ER lipid droplet compartment is promoted by the heteromeric complex formed by MTTP and the ER chaperone protein disulfide isomerase (PDI) (FIGURE 3). Studies where *Pdia1* was conditionally deleted in mouse hepatocytes greatly decreased VLDL secretion by inhibiting *Mttp* mRNA translation and led to APOB misfolding and degradation ¹²⁹. TM6SF2 is a transmembrane ER associated protein ¹³⁰ that promotes additional lipidation of nascent VLDL (Step 2, FIGURE 3), most likely working downstream of MTTP, and functions in formation of mature VLDL particles. A comprehensive study of subjects harboring inactivating mutations in *APOB* (familial hypobetalipoproteinemia, FHBL) or *MTTP* (abetalipoproteinemia, ABL) ³⁷ revealed profound hypolipidemia with reduced APOB, VLDL and LDL, combined with hepatic steatosis yet with varied progression of MASLD, with 20% homozygous FHBL and 10% ABL subjects developing advanced fibrosis³⁷. This heterogeneity in MASLD development in subjects with FHBL has been previously noted ¹³¹. Very recent findings from a cohort of patients with advanced MASLD revealed that subjects harboring rare *APOB* variants were predisposed to develop fibrosis and hepatocellular cancer and were protected against developing coronary artery disease¹³². Those recent findings reinforce the importance of the underlying pathobiology of hepatic steatosis in predicting outcomes of patients with MASLD and further suggest the opportunity for risk stratification in subjects identified as harboring *APOB* variants.

A loss-of-function variant in *TM6SF2* *rs58542926* was identified by several groups ^{133–135} who discovered the E167K variant was associated with hepatic steatosis and impaired VLDL triglyceride production. Further studies revealed that carriers of the minor *TM6SF2* allele exhibited reduced serum LDL cholesterol and reduced triglyceride levels and were protected from CVD ¹³⁶. In addition to impaired VLDL triglyceride secretion and hepatic steatosis, studies in a Finnish cohort of patients harboring the E167K *TM6SF2* variant demonstrated defective incorporation of polyunsaturated fatty acids into both liver and serum triglycerides, suggesting that alterations in hepatic phospholipid is also impaired and leads to a deficiency in phosphatidylcholine species containing polyunsaturated fatty acids ¹³⁷. The mechanisms underlying the observed shifts in polyunsaturated fatty acid channeling into phospholipid remain to be elucidated.

In addition, studies in mice and rats with germline *Tm6sf2* deletion revealed impaired hepatic VLDL triglyceride (but not ApoB) secretion ^{130,138}, which was independently confirmed in mice with conditional, liver-specific *Tm6sf2* deletion ¹³⁹. However, there remain unanswered questions regarding discrepant findings in studies examining *TM6SF2* loss-of-function phenotypes in humans. In particular, kinetic studies in subjects harboring the E167K variant demonstrated decreased secretion of both VLDL triglyceride and APOB100, suggesting that the *TM6SF2* variant modifies VLDL particle number and composition (ie decreases both APOB100 and triglyceride) rather than just decreasing VLDL lipidation with secretion of the same number of smaller, underlipidated particles (as in mice and rats) ^{140,141}. Recent studies using human induced pluripotent stem

cells, modified to express the *TM6SF2* E167K variant, demonstrated decreased APOB100 secretion yet found a paradoxical increase in intracellular APOB100¹⁴², findings at odds with other studies using human liver-derived spheroids where intracellular APOB100 was decreased¹⁴³. Accumulation of intracellular APOB in the setting of impaired secretion would be unanticipated based on existing models (reviewed in¹²⁸). While both studies suggest that human liver-derived models exhibit decreased VLDL particle number^{142,143} (as in the kinetic studies^{140,141}), important details regarding the fate of intracellular (human) APOB and the mechanisms accounting for the differences in rodent (mouse, rat) hepatic ApoB production^{130,138,139} will require further study (FIGURE 3).

Another area for further investigation concerns the possible loss-of-function in intestinal chylomicron formation and triglyceride transport and other phenotypes associated with the E167K *TM6SF2* variant. Studies from an Amish cohort demonstrated a subtle, albeit statistically significant, decrease in post-prandial serum triglyceride levels suggesting impaired intestinal lipid absorption, findings confirmed in fetal-like colon cells (Caco-2) and gene targeted zebrafish¹⁴⁴. Recent studies in intestine-specific *Tm6sf2* knockout mice demonstrated the development of spontaneous MASH, catalyzed by an altered microbial community and impaired barrier function¹⁴⁵. Those findings have yet to be replicated and the functional impact of the E167K variant in humans remains open. Recent reports of deep phenotyping of (>240,000) subjects screened for protein-altering variants of *TM6SF2* noted no digestive disorders, although subtle defects in intestinal lipid handling (insufficient to cause steatorrhea) may have been overlooked¹⁴⁶. In relation to the impact of gene variants involved in both hepatic VLDL and intestinal chylomicron secretion, it is worth noting that subjects harboring mutations in *APOB* or *MTTP* frequently manifest steatorrhea in association with intestinal lipid malabsorption³⁷. In addition, studies in mice with intestine-specific *Mttp* deletion exhibit impaired chylomicron secretion and adaptive changes in bile acid metabolism involving shifts in microbial taxa¹⁴⁷. Furthermore, disrupted intestinal barrier function in mice with intestine-specific *Mttp* deletion revealed compensatory increases in hepatic DNL and demonstrated impaired resolution of liver injury following high fat diet-induced MASH, despite up to 90-fold reductions in hepatic steatosis¹⁴⁸. Those studies suggest that loss-of-function variants that function in both hepatic and intestinal triglyceride mobilization and VLDL/chylomicron formation are associated with tissue-specific adaptations (including altered gut microbial composition and signaling) that may impact both fibrogenic signaling and lipoprotein metabolism.

Other genetic variants associated with VLDL assembly include **APOE** (FIGURE 2, 3) which is secreted from hepatocytes on VLDL particles and exchanges onto LDL particles for LDLR-dependent uptake. Recent studies from the UK Biobank found the *APOE rs429358* variant (encoding *APOE4*) was associated with parameters of liver damage and hepatic steatosis¹⁴⁹, with replication studies in a Finnish cohort showing this APOE variant associated with reduced steatosis and is protective against steatotic liver disease¹⁴⁹. Other studies identified the *APOE rs429358* variant to be associated with development of MASLD and the association clustered with body composition (BMI, obesity) measures¹⁵⁰. It is worth noting that APOE variants have been associated with both increased¹⁵⁰ and decreased²⁸ risk of CVD in subjects with MASLD (FIGURE 2, 3). Carriers of the APOE

minor allele exhibited increased circulating triglyceride and LDL cholesterol levels, which would be predicted to increase CVD risk in addition to its established role in Alzheimer's disease ¹⁴⁹. This same study also demonstrated that a gain-of-function variant in **GPAM** *rs2792751* (FIGURE 2, FIGURE 3) was associated with increased hepatic steatosis ¹⁴⁹. Earlier studies in mice demonstrated that overexpression of GPAM led to hepatic steatosis and increased VLDL triglyceride secretion ¹⁵¹ while germline *Gpam* deletion exhibited reduced hepatic lipid content when fed a high fat diet and improved glucose tolerance ¹⁵².

Earlier findings implicated a locus on human chromosome 8q24 containing **TRIB1**, which was associated with plasma levels of triglycerides and LDL cholesterol, where the minor allele was associated with protection against CVD (FIGURE 3) ¹⁵³. Those findings were confirmed in mice where overexpression of *Trib1* decreased hepatic VLDL triglyceride secretion and *Trib1* deletion demonstrated increased VLDL secretion and increased plasma cholesterol and triglyceride levels ¹⁵⁴. Those findings were further replicated in studies using human (HepG2) cells and provide support for a functional role of TRIB1 as a genetic modifier of VLDL secretion ¹⁵⁵, likely mediated through alterations in LDLR expression and alterations in DNL ¹⁵⁶. An intronic variant of TRIB1 *rs2980888* has been shown to modulate steatotic liver disease, although the mechanisms are as yet unresolved ¹⁵⁷.

Findings have also demonstrated an association with **TOR1B** *rs7029757* in MASLD linked to increased serum LDL cholesterol, hypertension and increased CVD (FIGURE 3) ¹⁵⁰, although the linkage to increased CVD was not confirmed in a recent study ²⁸. The mechanisms underlying these associations have been extrapolated from studies in mice where conditional, liver-specific deletion of the related gene *Tor1a* or its obligate partner *Lap1* demonstrated marked hepatic steatosis and impaired VLDL triglyceride secretion along with accumulation of nuclear lipid droplets ¹²⁴.

Somatic mutations in genes affecting VLDL secretion may arise in the course of the progression of MASLD to hepatocellular cancer, one prominent example being APOB where mutations were reported in up to 10% of subjects ¹⁵⁸. Recent studies have identified somatic mutations in **TBX3** that were proposed to enhance clonal expansion of hepatocytes in patients with MASLD and to function by augmenting cholesterol biogenesis and increasing VLDL secretion, independent of changes in APOB, MTTP or TM6SF2 ¹⁵⁹.

Genetic variants associated with MASLD, with primary impact on DNL or beta-oxidation

A variant in **GCKR** *rs1260326* that encodes a P446L substitution was associated with hepatic steatosis and increased circulating levels of triglycerides and LDL cholesterol (FIGURE 3) ¹⁶⁰. The P446L variant is a gain-of-function allele, as shown by in-vitro studies demonstrating enhanced glycolytic flux, promoting hepatic glucose metabolism and increased DNL, which together lead to increased secretion of hepatic VLDL, and increased serum triglyceride levels along with reduced levels of glucose ^{161,162}.

Genome-wide association studies have also identified a variant in **MBOAT7** *rs641738* in European subjects in association with alcohol related cirrhosis ¹⁶³ (FIGURE 2, 3) and more recent studies have implicated this *MBOAT7* variant in subjects with MASLD ¹⁶⁴. Functional studies in mice with targeted, liver-specific *Mboat7* deletion demonstrated

upregulation of DNL via Srebp1 as the primary defect leading to hepatic steatosis, without changes in hepatic VLDL triglyceride secretion ¹⁶⁵. Other studies demonstrated impaired lysosomal lipid homeostasis and altered autophagic flux in liver-specific *Mboat7* knockout mice may underlie the response to alcohol related injury ¹⁶⁶.

Studies in diverse ancestries and from different centers identified a variant in ***MARC1*** *rs26424238*, encoding an A165T substitution was associated with protection against MASLD (FIGURE 2, 3) ^{150,167}. Mechanistic studies, pursued in mice and in human liver-derived revealed the variant *MARC1* functions as a hypomorph ¹⁶⁷ and undergoes increased ubiquitylation with accelerated degradation along with mislocalization from the outer mitochondrial membrane ¹⁶⁸, suggesting the variant functions as a loss-of-function allele. Studies where *MARC1* was targeted in human liver cell lines (either using siRNA or CRISPR-mediated deletion) demonstrated increased fatty acid catabolism via beta oxidation pathways, which reduced triglyceride content, along with reduced ferroptosis and decreased levels of reactive oxygen species ¹⁶⁹. Those findings together provide support for considering therapeutic targeting of *MARC1* as a strategy in patients with MASLD.

Another target recently identified from a candidate gene approach revealed a variant in ***PSD3*** *rs71519934*, encoding an L186T substitution, which was protective against MASLD (FIGURE 3) ¹⁷⁰. *PSD3* expression (mRNA) was increased in liver biopsy samples from subjects with MASLD, and cultures of primary human hepatocytes from donors homozygous for the variant in *PSD3 rs71519934* exhibited reduced neutral lipid content, and siRNA knockdown of the WT allele in primary human hepatocytes demonstrated reduced steatosis ¹⁷⁰. These authors also demonstrated that *Psd3* knockdown in mouse hepatocytes fed a MASH-inducing diet led to a reduction in genes regulating DNL ¹⁷⁰. Further studies replicating these findings are awaited.

Genetic variant associated with MASLD, with unknown mechanism

A loss-of-function splice variant of the lipid droplet associated protein ***HSD17B13*** *rs72613567* was linked to protection against MASLD progression (FIGURE 3) and reduced fibrosis ¹⁷¹ and reduces the risk of both cirrhosis and of hepatocellular cancer development in subjects consuming at-risk quantities of alcohol ¹⁷². Another loss-of-function variant in *HSD17B13 rs62305726*, encoding a P260S substitution variant, was also linked to decreased MASH severity ¹⁷³. The functions of *HSD17B13* include a role as a retinol dehydrogenase ¹⁷³ as well activities in steroid and eicosanoid metabolism, shifts in phospholipid metabolism ¹⁷⁴ and alterations in pyrimidine catabolism ¹⁷⁵. However, studies in mouse models examining the mechanisms and pathways of how this loss-of-function allele may confer protection against MASH and fibrotic injury have yielded conflicting results. Germline deletion of *Hsd17b13* revealed no protection against high fat diet-induced MASH or hepatocellular cancer ¹⁷⁶, while another report showed antisense *Hsd17b13* oligonucleotide knockdown in mice decreased hepatic steatosis but did not affect fibrosis ¹⁷⁷. In an attempt to reconcile these differences, recent studies generated a new *Hsd17b13* knockout and found only minimal differences in liver inflammatory markers or fibrotic injury, concluding that species differences between mice and humans likely obscure the role of HSD17B13 in human liver injury ¹⁷⁸. Nevertheless, and despite the challenges

in assigning possible mechanisms by which the loss-of-function *HSD17B13* allele might confer protection in subjects with MASH, a recent phase 1 study of an siRNA (Rapirosiran) demonstrated a dose-dependent reduction in liver HSD17B13 mRNA abundance and revealed reduced NAS activity and fibrosis scores at 6 and 12 months in 44 subjects with MASH (Fibrosis grade 1-2), and no changes in steatosis¹⁷⁹. Recent work has implicated a role for HSD17B13 in the pathogenesis of hepatocellular cancer through altered metabolic signaling mediated via an ER protein EMC6¹⁸⁰, whose targeted deletion was accompanied by microvesicular lipid droplet accumulation¹⁸⁰.

Targeting modifiers of hepatic VLDL metabolism independent of MASLD

APOC3, whose circulating levels are predictive of CVD, functions as an inhibitor of VLDL lipolysis and impairs TRL remnant particle uptake (FIGURE 2)¹⁸¹. Studies evaluating APOC3 as a CVD risk and Mendelian randomization studies each support the clinical development of APOC3 as a CVD therapeutic target^{182–184}. Silencing ApoC3 in mice prevents atherosclerotic lesion development^{185–189}, and therapeutic targeting of APOC3 reduced serum triglyceride levels, without aggravating hepatic steatosis and with no change in VLDL production^{190–195}. The emerging consensus suggests the protective effects of APOC3 silencing are mediated by reducing levels of triglyceride-rich particles (including VLDL remnants) rather than by reductions in LDL-cholesterol levels¹⁹⁶. The impact of APOC3 silencing on pathways relevant to MASLD progression are yet to be fully elucidated.

Studies of individuals with low levels of circulating plasma LDL cholesterol and triglycerides implicated *ANGPTL3* as a causal locus and further demonstrated *ANGPTL3* null individuals were protected against CVD^{197,198}. Antisense inhibitors of *ANGPTL3* effectively reduce LDL cholesterol levels even in the absence of LDLR expression¹⁹⁹. *ANGPTL3* functions as an inhibitor of VLDL lipolysis (FIGURE 3), through interactions with peripherally acting lipases²⁰⁰. Other studies suggest that *ANGPTL3* deficiency modulates VLDL production, but with conflicting results (including species differences between mice and humans) regarding APOB100 secretion and hepatic neutral lipid accumulation^{201–204}. Recent studies in subjects with compound heterozygous *ANGPTL3* loss-of-function observed reduced hepatic secretion of triglyceride-rich particles, coupled with increased plasma clearance of both gut and liver-derived triglyceride-rich particles²⁰⁵. However, it remains possible that the extremely rapid hepatic uptake of these triglyceride-rich particles may possibly obscure differences in APOB production, leaving open the question of whether the effects of *ANGPTL3* silencing are confined to VLDL clearance. Recent phase 2b studies of the *ANGPTL3* silencing RNA (Zodasiran) in patients with mixed hyperlipidemia revealed a subset of subjects (most of whom were obese with BMI>31) who exhibited increased hepatic steatosis at baseline (using MRI-PDFF imaging), where siRNA treatment decreased liver fat (–28% vs –2% in patients receiving placebo) at 24 weeks²⁰⁶, raising the possibility that targeting *ANGPTL3* may directly impact hepatic lipid homeostasis. In addition, recent studies demonstrated that shRNA silencing of *Angptl3* in mice decreased hepatic fructose uptake through mechanisms including reduced GLUT8 expression, suggesting that *ANGPTL3* may exert effects on hepatic pathways involved in energy metabolism²⁰⁷.

CONCLUSIONS

CVD is among the leading causes of morbidity and mortality in subsets of patients with MASLD, whose basis reflects alterations in the production and clearance of triglyceride-rich lipoproteins (including VLDL) and LDL. Further refinement in deep phenotyping, coupled with advances in metabolomic and lipidomic profiling will further strengthen understanding of how genetic and environmental modifiers influence MASLD progression and the development of adverse cardiometabolic outcomes, including CVD. Moving forward, developing targeted therapeutics that mitigate risk of CVD and improve cardiometabolic outcomes in MASLD patients will require an interdisciplinary approach to understand the complex interactions at play.

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SUMMARY

- Atherosclerotic and cardiometabolic disease is a critical clinical problem faced by patients with MASLD.
- Emerging information points to distinct metabolic and genetic subsets of MASLD
- Understanding pathways of hepatic lipoprotein production may reveal new targets for mitigating cardiometabolic disease in patients with MASLD.

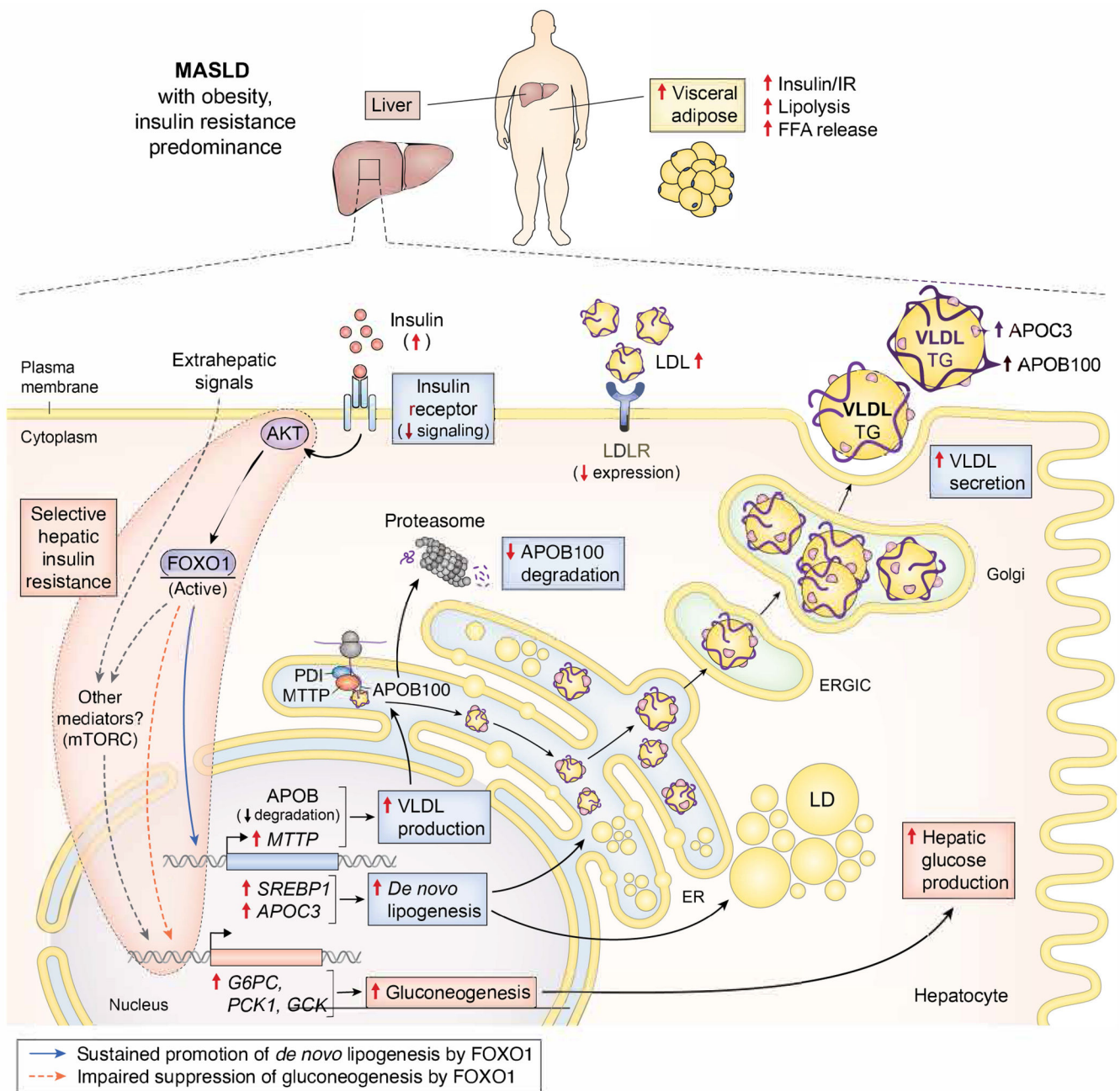


Figure 1. Metabolic pathways contributing to dyslipidemia and aberrant hepatic lipoprotein metabolism in subjects with MASLD accompanied by insulin resistance or type 2 diabetes. Obese subjects accumulate visceral adipose depots which release increase quantities of free fatty acids (FFA), as a result of systemic insulin resistance and hyperinsulinemia which leads to impaired suppression of adipose triglyceride lipolysis. In addition, increased circulating glucose levels, along with consumption of fructose and hyperinsulinemia together promote increased hepatic de novo lipogenesis (DNL). Those features of insulin signaling are further modified under conditions of selective hepatic insulin resistance in MASLD, where impaired insulin receptor signaling, coupled with additional (extrahepatic) signaling results in continued (paradoxical) transcriptional upregulation of gluconeogenesis

and increased hepatic glucose production. In addition, there is an adaptive increase in downstream pathways including AKT activation which in turn leads to impaired suppression of FOXO1 activity which results in transcriptional activation of downstream targets and master regulators of DNL (SREBP1) and VLDL production (MTTP). These adaptations result in increased production of hepatic VLDL triglyceride. Hepatic VLDL production is further enhanced by FOXO1-dependent pathways that reduce co-translational degradation of APOB100, which leads to production of increased numbers of APOB100-containing VLDL particles, each particle carrying additional neutral lipid, along with increased APOC3. The increased production of VLDL particles eventually results in increased LDL, which accumulate in plasma because of downregulation of LDLR expression secondary to hepatic insulin resistance.

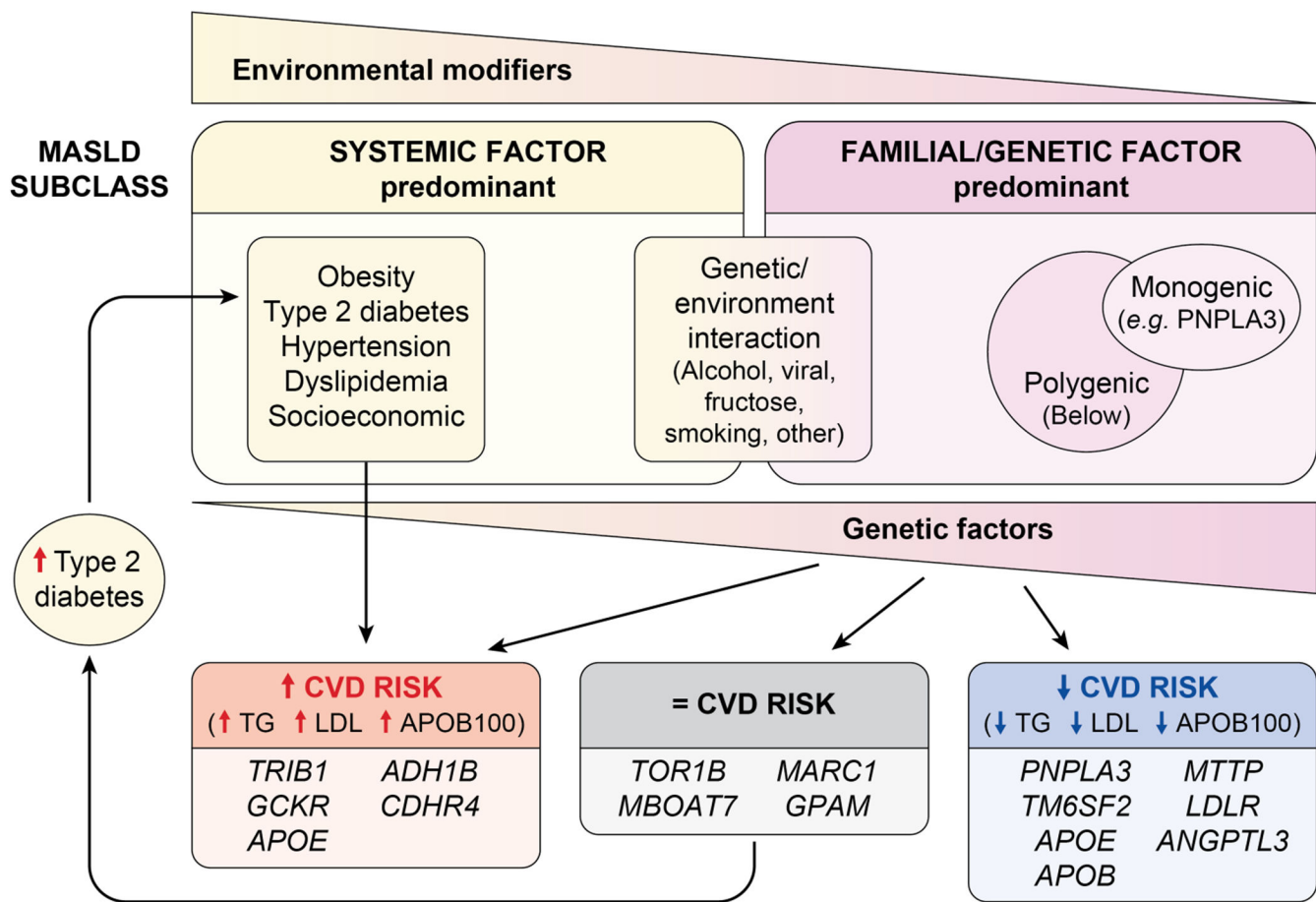


Figure 2. Proposed classification of MASLD subtypes segregated by systemic factor predominance versus familial/genetic factor predominance, illustrating the role of genetic/environmental interactions.

MASLD exhibits striking heterogeneity in underlying etiologies and in the rate of progression and the development of comorbid outcomes (including cardiovascular disease, CVD). Emerging evidence from deep metabolic and clinical phenotyping, coupled with developments in polygenic risk score assignment have allowed a predictive gradation of disease progression that align with two predominant subtypes of MASLD. Studies combining metabolomic and lipidomic profiling, coupled with increased understanding of genetic factors (monogenic and polygenic) have identified subsets of patients with familial/genetic factor predominant MASLD, where decreased plasma triglycerides, APOB100 and LDL cholesterol levels align with reduced risk of CVD. A second subset of systemic factor predominant MASLD includes subjects at increased CVD risk that aligns with the presence of obesity, type 2 diabetes, hypertension and dyslipidemia. In this systemic factor predominant MASLD subset, there are genetic factors that further increase CVD risk. These subsets are not mutually exclusive and environmental modifiers interact with genetic susceptibility to modify outcomes and comorbidities. In addition, genetic modifiers of obesity, hypertension and type 2 diabetes function independently and in combination within the subtypes outlined.

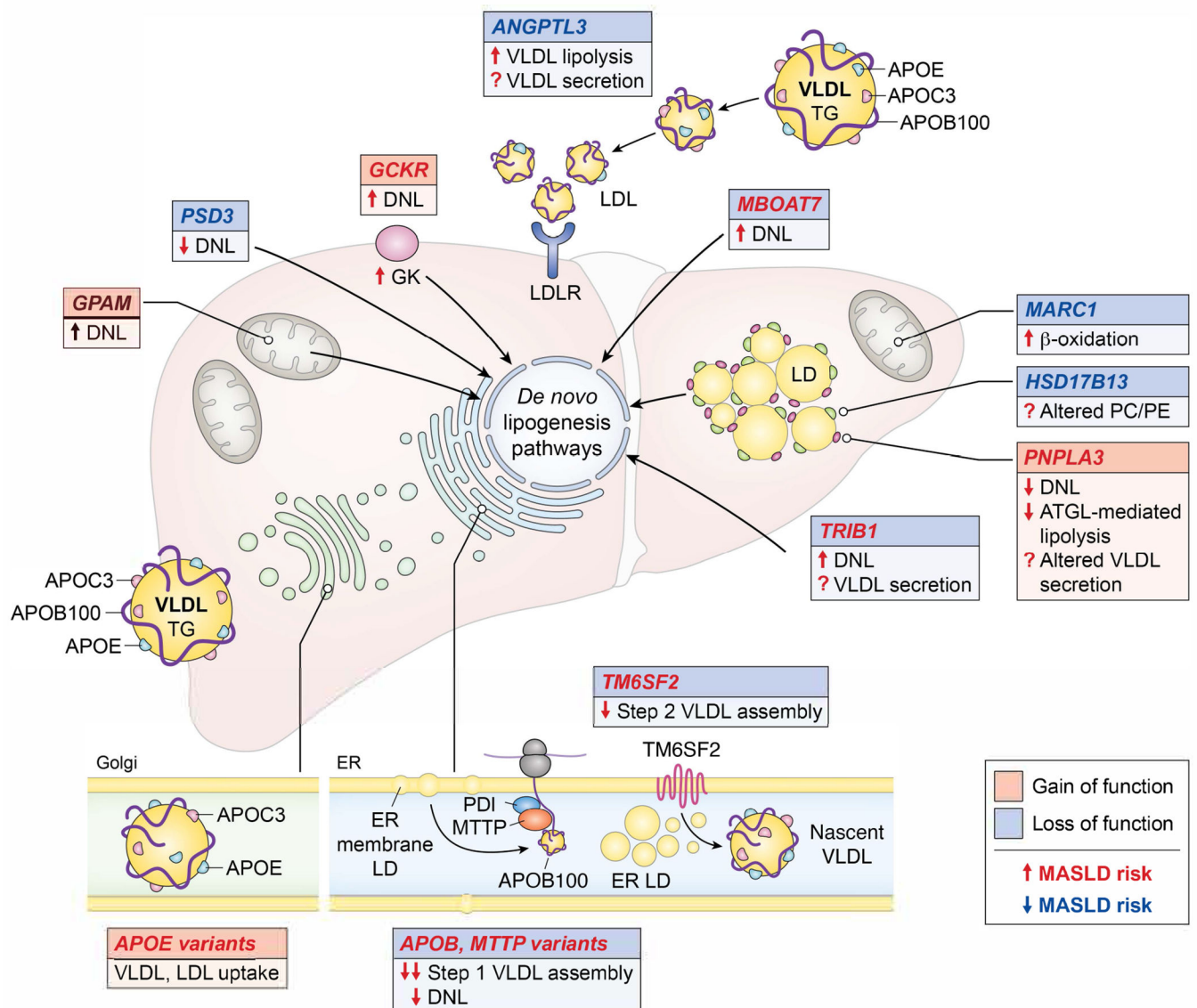


Figure 3. Genetic variants linked to MASLD and role in pathways including de novo lipogenesis, lipid droplet metabolism, mitochondrial function/energy utilization, and VLDL assembly/secretion.

Genetic variants associated with increased risk of MASLD are indicated in bold red; variants associated with decreased MASLD risk are indicated in bold blue. *PNPLA3*, the most prevalent risk allele for MASLD, is a lipid droplet (LD) associated protein and likely functions as a gain-of-function allele in inhibiting adipose triglyceride lipase (ATGL)-mediated lipolysis and suppression of de novo lipogenesis (DNL). There is controversy as to a role for the *PNPLA3* variant in altering VLDL production. Two loss-of-function variants have been described for *HSD17B13* in association with protection against progressive forms of MASLD. The underlying mechanisms for the protective effect of the variant *HSD17B13* alleles are incompletely understood, attributable in part to divergent functions in mice versus human lipid droplets, but a role for altered phospholipid composition and impaired pyrimidine metabolism have each been invoked (see text). A loss-of-function variant in

MARC1 encodes an unstable protein, decreasing hepatic steatosis through upregulation of mitochondrial beta oxidation. A variant in *MBOAT7* promotes hepatic steatosis by increasing DNL (see text). A variant in *GCKR* increases hepatic steatosis in part by increasing glucokinase (GK) activity and channeling glucose for triglyceride production. A variant in *PSD3* demonstrated reduced DNL in mice as a mechanism for protection against MASLD. A variant in GPAM manifests a gain-of-function in promoting hepatic triglyceride production through increased DNL. Variants in *APOB*, *MTTP* and *TM6SF2* function to impair key steps in the assembly of VLDL particles in the endoplasmic reticulum (ER). MTTP functions in parallel with an ER chaperone protein, protein disulfide isomerase (PDI) in mobilizing membrane bound and intraluminal ER lipid droplets. TM6SF2 functions in a second step of VLDL assembly to further promote lipidation of nascent VLDL. Variants in *TRIB1* have been implicated in promoting MASLD, through upregulation of DNL and possible alterations in VLDL secretion. Variants in *ANGPTL3* have been linked to decreased hepatic steatosis and function in part by increasing peripheral lipolysis of VLDL particles and possibly also by altering hepatic VLDL assembly and secretion. *APOE* variants have been linked to increased MASLD risk through unknown mechanisms, but these variants exhibit functions in VLDL and LDL particle uptake.

Table 1.
Genetic modifiers of hepatic VLDL assembly and secretion

Gene	Mechanism of action	Reference
APOB	Initiation of primordial VLDL particle; step 1	Within text
MTTP	Lipidation of nascent APOB within ER; step 1	Within text
PDI1A	Obligate chaperone for MTTP	Within text
TM6SF2	Maturation and continued lipidation; step 2	Within text
miR130b	Increases MTTP expression, VLDL secretion	Zhang et al
SMLR1	MTTP function in VLDL secretion; ?mechanism	van Zwol et al
LAP1	Defective APOB translocation, lipidation	Within text
PLA2G12B	Lipid channeling into ER; VLDL initiation	Thierer et al
TMEM41B/VMP1	ER scramblase; PL transfer into ER	Morishita et al; Huang et al; Jiang et al 2022
LPCAT3	Arachidonyl PL production in VLDL assembly	Rong et al
GSL1	Increases PL flux, promotes VLDL assembly	Simon et al
SMS1/SMS2	Sphingomyelin production in VLDL assembly	Li et al
tPA	Modifies MTTP-APOB interaction	Dai et al
DGAT1/DGAT2	TG availability for VLDL assembly	Irshad et al; Amin et al; Calle et al; Loomba 2025
DIESL/TMEM68	DGAT-independent TG for VLDL assembly	McLelland et al
SAR1B	COPII ER-Golgi VLDL transport	Gusarova et al; Siddigui, Tiwari et al
SURF4	SAR1B-dependent VLDL exit	Wang B et al; Wang X et al